

Reflections on Educational Game Prototyping and Agile Iteration Based on the Agile Scrum Framework Report

Section 1: Personal contributions to the team design portfolio

User story 1 Name:

Stub: Enter the color laboratory

User story: As a kindergarten student. I want to click a large animated start button so that I can easily enter the game without reading any text.

Acceptance criteria: The home screen has a vibrant cartoon background with a bouncing button. animation.

Clicking the button plays a magic sound and transitions to the core scene. scene.

Story size: M

URL to prototype (Figma prototype design):

<https://www.figma.com/board/wcrZx8euXC17OJRNFkbiT/Untitled?node-id=1-5&t=RB35blqDcUiyjX0D-1>

URL to the game: <https://www.kopo.com/?sid=2972&lid=5378>

Sprint Pilot Backlog

ID	As a	I want to able to play	So that
1	Child	Fruit Sorting Game	Learn how to classify fruits by their colors while having fun
2	Child	Drag and Drop Game	Improve hand-eye coordination by dragging fruits to target areas
3	Child	Fruit Recognition Game	Recognize different types of fruits and their natural colors
4			
5			
6			

User story 2 Name:

Stub: Letter Sorting Laboratory / Alphabet Matching

User story: As a kindergarten student, I want to drag object cards into the corresponding letter columns so that I can practice phonics and word recognition through play.

Acceptance criteria:

The screen displays four letter columns (A, C, D, B) with distinct colorful frames. Bottom section displays draggable item icons (e.g., apple, cat, dog, plane, car, bee). Dragging an item into the correct letter card places it inside, updating the score at the top right.

Top header shows current game level ("Level 2") and pause/profile status.

Story size: M

URL to prototype (Figma prototype design):

<https://www.figma.com/board/wcrZx8euXC17OJRNfKbkiT/Untitled?node-id=1-5&t=RB35blqDcUiyjX0D-1>

URL to the game: <https://www.kopo.com/?sid=2972&lid=5378>

ID	As a	I want to able to play	So that
1	Child	Letter Sorting Game	Learn how to classify items by their starting letters while having fun
2	Child	Drag and Drop Game	Improve hand-eye coordination by dragging item cards to target letter columns
3	Child	Object & Phonics Recognition Game	Recognize different objects (e.g., apple, dog, car) and connect them with their initial letters
4			
5			
6			

User story 3 Name:

Stub: Color Sorting Laboratory / Color Matching

User story: As a kindergarten student, I want to drag colorful items into their matching color containers so that I can learn and reinforce color recognition while playing.

Acceptance criteria:

The screen displays four color columns (BLUE, GREEN, RED, YELLOW) with corresponding colored borders and labels.

The bottom section provides various draggable objects (e.g., olive, tree, whale, flower, car, chick, cherries, bee, pepper, grapes).

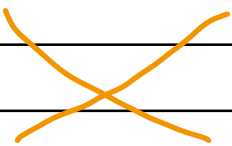
Dragging an item into its corresponding color column places it inside and updates the score display in the top right.

Top header indicates game level ("Level 3") with prompt ("Let's categorize them by color!") and controls for pause/profile.

Story size: M



ID	As a	I want to able to play	So that
1	Child	Color Sorting Game	Learn how to classify items by their dominant colors while having fun
2	Child	Drag and Drop Game	Improve hand-eye coordination by dragging colored objects into target color columns
3	Child	Visual & Color Recognition Game	Identify various everyday objects (e.g., cherries, tree, grapes) and match them with correct color names
4			
5			
6			



URL to prototype (Figma prototype design):

<https://www.figma.com/board/wcrZx8euXC17OJRNfkbkiT/Untitled?node-id=1-5&t=RB35blqDcUiyjX0D-1>

URL to the game: <https://www.kopo.com/?sid=2972&lid=5378>

Section 2: Rationale behind key design decisions you were involved in

2. Project objectives, deliverables, and rationale for prototype design

In Agile development practices, the Scrum framework emphasizes rapid value delivery in environments of uncertainty through collaboration between self-managing and cross-functional teams. Following in-depth customer interviews, our team clarified the prototyping requirements and constraints for the children's educational game (Beck et al., 2001).

2.1 Core Objectives, Deliverables, and Constraints

Firstly, the client wanted to develop an educational game prototype that was fun, engaging, motivating, and exciting, covering language, cognitive, and scientific fundamentals (Norman, 2013). The client specifically requested that three game levels and three corresponding game themes be designed for each student. I chose the sorting game.

Secondly, the deliverables include interactive prototype increments for three core levels, conforming to a predefined Definition of Done (Praise for Essential Scrum, n.d.).

Thirdly, the time constraint is limited to a fixed-length Sprint cycle, and the team must prioritize delivering the MVP (Minimum Viable Product) within a limited time.

Fourthly, the assumptions are that children can intuitively understand categorization logic through drag-and-drop matching with visual colors/initials, and that the game interface feedback is sufficient to maintain their focus.

2.2 Requirements refinement and agile prioritization

Based on customer needs, our team broke down the Product Backlog into specific user stories and developed acceptance criteria (Schwaber & Sutherland, 2020).

User story 1 (Level 1: Fruit Color Classification) Acceptance Criteria: Players can correctly drag and categorize the fruits of various colors below into the four areas: GREEN, RED, YELLOW, and BLUE. User story 2 (Level 2: First Letter Sorting) Acceptance Criteria: Players can identify objects (such as Cat, Donut) and match them into the corresponding English letter A, B, C, D containers. And user story 3 (Level 3: Cross-Category Color Abstract Classification) Acceptance Criteria: Players can classify mixed items such as animals, plants, and vehicles into color containers according to their main color.

2.3 Game Prototype and Asset Design Analysis

To meet the acceptance criteria of "Engaging, Motivating, Exciting, Fun", our team adopted targeted UX/UI design strategies in the three stages (Norman, 2013).

First is Level 1, Fruit Color Sorting. The interface uses bright, highly saturated colors (rainbow, green background) to suit children's visual preferences. Below are images of real fruits such as bananas, apples, and grapes, with clear English color labels (GREEN, RED, YELLOW, BLUE) for the target containers. This design reduces initial cognitive load, allowing children to quickly get started and receive immediate feedback and motivation.

Next is Level 2, sorting the first letters of English words. This level elevates the challenge to language recognition, requiring participants to drag items, such as Cat, Apple, Book, Car, into their corresponding letter containers (A, B, C, D). Card-style artistic borders and cartoon illustrations enhance the fun and sense of exploration.

Finally is Level 3, cross-category color categorization. This level increases the challenge by mixing animals (whales, chicks), vehicles (cars), and plants (trees, peppers). The design incorporates distractors and counterintuitive challenges, such as yellow chicks and red cars. This encourages children to break free from fixed thinking patterns and categorize colors based on their inherent nature. It significantly enhances the engaging aspect of the game and the excitement of completing the level.

Section 3: Reflects on feedback, iteration, constraints, and design trade-offs

3.1 Process Review and Team and Client Feedback

During the agile pilot Sprint, the team explored the technical feasibility of drag-and-drop interaction through Spike. However, during the first team meeting and Daily Scrum, our team members felt significant assessment and time pressure. After observing the initial prototype, our team expressed concerns about balancing the game's "learning effectiveness" and "entertainment." Some felt it was too difficult for 4- to 6-year-old kindergarten students. Our team briefly disagreed on the balance between Outputs and Outcomes.

3.2 Performance and target achievement evaluation

The team conducted a self-assessment against the three pillars of transparency, inspection, and adaptation.

In terms of deliverables achievement, we have successfully completed three working level prototypes that meet the planned DoD. Regarding user engagement, playtesting showed that children completed Level 1 in an average of 20 seconds and attempted Level 3 an average of 2.3 times, indicating that the level progression design successfully maintained player focus. Regarding the deviation analysis, the initial focus on the presentation on Figam led to a reduction in the production time on Koop, resulting in some images still having unremoved white borders.

3.3 In-depth results analysis and design trade-offs

During the Sprint Review and Sprint Retrospective meetings, our team conducted an in-depth analysis of the results. In Agile development, it is essential to distinguish between outputs and outcomes. Outputs refer to the number of lines of code delivered, the number of pictures drawn, or Story Points, while Outcomes refer to whether the game truly leads to behavioral changes and learning value in children.

The team originally planned to design dynamic presentations for each subject topic, but after evaluating time constraints and Spike feedback, they made a design compromise. I unified the UI logic by changing different images. This compromise, while reducing the richness of visual outputs, ensures the rigor of the game logic and the smoothness of the learning experience.

3.4 Summary of core experiences and future action plan

Based on our team's core experience, we believe that the key is to learn before building and to give equal importance to process learning and design learning. When facing uncertain UX design, validating fragile assumptions through prototyping can significantly reduce ineffective development. Furthermore, Retrospective focuses on how the team collaborates (HOW), while Review focuses on what was built (WHAT); both are indispensable.

Going forward, my team and I should implement actionable improvements. In the next Sprint, we will clearly define and implement at least one specific improvement. For example, we will strictly limit Daily Scrum to 15 minutes and focus solely on blockers and collaborative planning, avoiding turning it into a mere progress report to the Scrum Master. On the other hand, we should also deepen our reflection. We will continuously record reflective evidence in the Team Design Portfolio and Individual Workbook, using the 5 Whys method to explore the root causes of problems and completely eliminate superficial reflection.

Reference :

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